

Math 526, S05
Quiz 6

Name:
SSN: xxx-xx-

Instructions: There are totally 2 problems. You must show all of your work to receive a credit for a correct answer. No graphical calculator is allowed in the quiz and exam.

Problem 1. (12 points) Let $\mathbf{A} = \begin{bmatrix} 2 & 3 \\ 3 & 5 \end{bmatrix}$. Find $\kappa_1(\mathbf{A})$ and $\kappa_\infty(\mathbf{A})$.

Solution: Using the formula in Exercise 6.4, it is easy to find that

$$\mathbf{A}^{-1} = \begin{bmatrix} 5 & -3 \\ -3 & 2 \end{bmatrix}$$

Then, we have

$$\begin{aligned} \|\mathbf{A}\|_1 &= \max\{2+3, 3+5\} = 8, & \|\mathbf{A}^{-1}\|_1 &= \{5+|-3|, |-3|+2\} = 8, \\ \|\mathbf{A}\|_\infty &= \max\{2+3, 3+5\} = 8, & \|\mathbf{A}^{-1}\|_\infty &= \{5+|-3|, |-3|+2\} = 8. \end{aligned}$$

So

$$\begin{aligned} \kappa_1(A) &= \|\mathbf{A}\|_1 \|\mathbf{A}^{-1}\|_1 = 8 \times 8 = 64 \\ \kappa_\infty(A) &= \|\mathbf{A}\|_\infty \|\mathbf{A}^{-1}\|_\infty = 8 \times 8 = 64 \end{aligned}$$

Problem 2. (8 points) How many significant digits does the approximation $x' = 0.0225$ of $x = 0.0296$ have?

Solution: Since

$$x = 2.96 \times 10^{-2}, \quad x' = 2.25 \times 10^{-2},$$

we have

$$0.5 \times 10^{-2} \leq |x - x'| = 0.71 \times 10^{-2} \leq 5 \times 10^{-2}$$

So this approximation has 0 significant digit.